



Competent  
 $\frac{40}{40}$   
100%

**ELECTRICITY AND ELECTRONICS  
FORMATIVE ASSESSMENT**

|                              |                                                  |
|------------------------------|--------------------------------------------------|
| Module #                     | 671101000-KM-03:                                 |
| NQF Level                    | level 4,                                         |
| Notional hours               | 130                                              |
| Credit(s)                    | Cr 13                                            |
| Occupation Specialisation or | 671101000: Occupational Certificate: Electrician |
| Associated Occupation        | 671101: Electrician                              |

### About the Formative Assessment

|                                     |                                                                                                                                                                                                                                                                                                                                                                     |
|-------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Purpose</b>                      | The purpose of this Formative Assessment is to evaluate learners understanding on the specific outcomes and/or assessment criteria and prepare the learner for the summative assessment. This will show that the learner is ready to be assessed.                                                                                                                   |
| <b>Context</b>                      | This assessment represents the Formative Assessment component and should be completed in the classroom/training room.                                                                                                                                                                                                                                               |
| <b>Resources</b>                    | The following are resources needed for this assessment:<br><ol style="list-style-type: none"> <li>1. Learner Guide; and</li> <li>2. Assessment Preparation.</li> </ol>                                                                                                                                                                                              |
| <b>Instructions to Facilitators</b> | Facilitators will be required to: <ul style="list-style-type: none"> <li>• Explain the completion of the Formative Assessment to each learner; and</li> <li>• Interview the learner on similar questions, should he/she not be able to write.</li> </ul>                                                                                                            |
| <b>Instructions to Learners</b>     | Learners will be required to: <ul style="list-style-type: none"> <li>• Complete the Formative Assessment as per the instructions;</li> <li>• Ensure that all questions are completed;</li> <li>• Ensure that the completion of the workbook is their own work;</li> <li>• Ensure that all annexure are attached to the workbook and clearly referred to;</li> </ul> |
| <b>Assessment Time</b>              | Learners are required to complete this assessment within the allocated time frame/hours.                                                                                                                                                                                                                                                                            |
| <b>Total Mark</b>                   | This formative assessment carries a total mark of <b>_ points</b> . In order to meet the pass mark, learners are required to achieve a minimum of <b>80%</b> of the total marks.                                                                                                                                                                                    |

### Special arrangements for assessment

Place Eastc

Language English

Resources

Barriers NA

### People to be involved with assessment

Learner: Levato

Trainer: R Nyakala

Facilitator

Assessor:

Manager:

Mentor / Coach:

Moderator:

### Next steps for learning

|  |
|--|
|  |
|  |
|  |

### Resources required for this assessment

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|  |

### Guidelines to the learner:

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|--|
|  |
|  |
|  |
|  |
|  |

Learners Name: Levato m DitsheGo

Learner's signature: [Signature]

Date: \_\_\_\_\_

Facilitator Name: R Nyakala

Facilitator's signature: \_\_\_\_\_

Date: \_\_\_\_\_

## ASSESSMENT APPEAL APPLICATION FORM

### LEARNER'S NOTICE OF ASSESSMENT APPEAL

**TO: The Internal Moderator**

A meeting with the internal moderator is hereby requested to discuss the outcome of my assessment.

Internal moderator name \_\_\_\_\_

Date of submission: \_\_\_\_\_

Name of employee assessed: \_\_\_\_\_

Name of Assessor: \_\_\_\_\_

Date of feedback session: \_\_\_\_\_

### Grounds for Appeal

| No | Tick the applicable ground(s) for appeal                                                                             | Tick |
|----|----------------------------------------------------------------------------------------------------------------------|------|
| 1  | The assessment did not follow the laid down procedure                                                                |      |
| 2  | Not all evidence available was taken into account during the assessment                                              |      |
| 3  | The assessor was not a subject matter expert nor was a subject matter expert present during the assessment process   |      |
| 4  | The assessment was not according to the performance criteria and the range statement stipulated in the unit standard |      |
| 5  | Not all the range items were available for the assessment                                                            |      |

## Reasons for Appeal

| No | Please give detailed reasons for the choice(s) above |
|----|------------------------------------------------------|
| 1  |                                                      |
| 2  |                                                      |
| 3  |                                                      |
| 4  |                                                      |
| 5  |                                                      |

Learner's signature: hDyru Co

Date: \_\_\_\_\_

Employee witness: \_\_\_\_\_

Date: \_\_\_\_\_



|                                                                                               |                            |
|-----------------------------------------------------------------------------------------------|----------------------------|
| Learner declaration of acceptance of assessment instruments and relevant documentation: Date: |                            |
| Learners Name:<br><i>herato</i>                                                               | Signature<br><i>Herato</i> |
| Facilitator Name:                                                                             | Signature:                 |



**LEARNING UNIT 1:**  
**KM-03-KT01: Fundamentals of electricity (35%)**

|                     |                                                                                               |
|---------------------|-----------------------------------------------------------------------------------------------|
| Learning Outcome 1: | Principles and fundamental concepts of electricity                                            |
| Learning Outcome 2: | Definitions, types, properties and applications of conductors, insulators and semi-conductors |
| Learning Outcome 3: | Concepts, theories and principles of Electrical Circuits                                      |
| Learning Outcome 4: | Calculations on basic electrical circuits (resistance, voltage and current) and power         |
| Learning Outcome 5: | Basic principles and calculation of magnetism                                                 |

| Task | Questions Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Marks |
|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| 1.   | <p>a. Describe, calculate and interpret fundamental concepts of electricity (Electro motive force, Potential difference, Resistance) using the correct units of measurement and definitions.</p> <p>b. Describe the factors that influences the resistance of a material</p> <p>c. Define and explain, using the correct units of measurement ohms law of electricity</p> <p>d. Define and explain using the correct units of measurement Kirchoff's law of electricity Manipulate formula to calculate voltage, current and resistance in series/parallel circuits</p> | 10    |

① (EMF) - Electro motive Force Unit (V)

- Potential- (voltage) is the difference in electrical potential between two points in a circuit

- Resistance - is the opposition to the flow of electricity current in a circuit.

- Resistance is measured in Ohm's Law ( $\Omega$ ) ( $R = V/I$ )

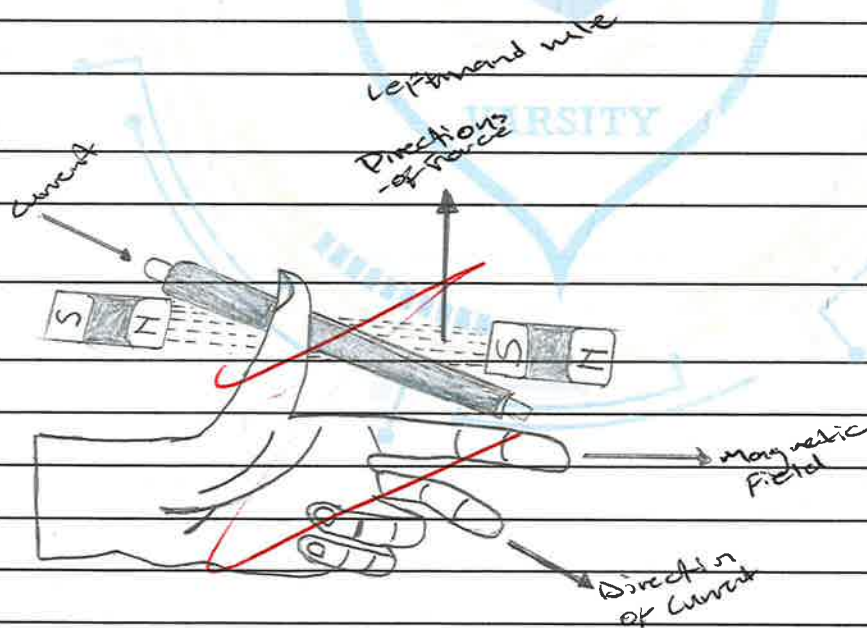
Voltage (V) = Current (I) x Resistance (R)

| Task | Questions Description                                                                                                                                                                                                                                                                                                                                       | Marks |
|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| 2    | <p>a) Describe, with the aid of drawings, the effect on the magnetic field around a current carrying conductor when placed in a uniform magnetic field. Fleming's left-hand rule must be demonstrated.</p> <p>b) Describe and explain the various forces or methods that can be used to alter magnetic fields, and describe the changes that take place</p> | 10    |

~~Parallel Circuit - the voltage is the same across all components in a parallel circuit~~

~~- Current: the total current is the sum of the individual current through each branch,  $I_{total} = I_1 + I_2 + I_3 + \dots$~~

~~- Resistance the reciprocal of the total resistance is the sum of the reciprocal of the individual resistors  $\frac{1}{R_{total}} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \dots$~~



## LEARNING UNIT 2:

### KM-03-KT03: Earthing and bonding (35%)

|                     |                                                                                             |
|---------------------|---------------------------------------------------------------------------------------------|
| Learning Outcome 1: | KT0301 Principles of earthing and bonding                                                   |
| Learning Outcome 2: | KT0302 Methods of earthing and bonding on low voltage overhead lines, equipment and cables. |
| Learning Outcome 3: | KT0303 Regulatory and statutory requirements related to earthing and bonding                |

| Task | Questions Description                                                                                                                                                                                                                                                                                                                           | Marks |
|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| 3    | <p>a) Explain the regulatory requirements for the earthing of neutral conductors on both the supplier and consumer side of an installation</p> <p>b) Explain the term 'systems earthing'</p> <p>c) Describe the earthing systems relevant to SANS 10142-1.</p> <p>d) Describe how a common earth electrode is used in reticulation circuits</p> | 10    |

Regulatory requirements for earthing of neutral conductors on both the supplier and consumers side, are designed to ensure safety and prevent electric shock

- Supplier side

\* the supplier neutral must be earthed at the supply transformer

\* Multiple earthing point might be required in the distribution network to maintain neutral potential close to earth potential

\* Regulation dictate the maximum earth loop impedance to ensure protective devices operate quickly in the event of a fault

TN-C-S - A system where the neutral and earth functions are combined in a single conductor (PEN) throughout the system. This system is generally not allowed for new installations in South Africa due to safety concerns.

TN-C-S - A system where the neutral and earth functions are combined in a single conductor (PEN) in part of the system, typically from the supply transformer to a point near the consumer's premises, and then separate into distinct neutral and conductors.

TT : A system where the supply transformer is earthed directly, and the consumer's installation has its own independent earth electrode.

IT : A system where the supply transformer is either isolated from earth or earthed through a high impedance. The consumer's installation has its own earth electrode.

↳ A common earth electrode in reticulation circuits is used to provide a shared earthing point for multiple consumers or installations. This is typically done in situations where it is difficult or impractical to install individual earth electrodes for each consumer.

The common earth electrode must have a low resistance to ensure effective fault current flow and must be adequately sized to handle potential fault currents. It is connected to the neutral conductor at multiple points in reticulation circuit to maintain a stable earth potential.

② Electrolytic Capacitors: These capacitors have a higher capacitance value compared to ceramic capacitors. They are polarized, meaning they have a positive and negative terminal.

Types

- Aluminum electrolytic capacitors: common and cost effective for general-purpose applications
- Tantalum electrolytic capacitors: offer better stability and higher temperature performance but are more expensive
- Applications
  - Power supply: used for smoothing rectified A-C voltage and energy storage
  - Audio Amplifiers: used for coupling and decoupling signal
  - Timing Circuits: used in app

EB charts are used to convert these codes into capacitance value in farads (F) microfarads (μF) picofarads (pF)

Capacitors in parallel

Total capacitor is the sum of all capacitors

$$C_T = C_1 + C_2 + C_3$$

Capacitors in series

$$\frac{1}{C_T} = \frac{1}{C_1} + \frac{1}{C_2}$$

c Resistance value are determined by colour bands on the resistance. Each colour represents a number on the colour represents a number on the colour chart the first band are numbers, the third band multipliers and fourth band tolerance value is calculated using the colour code chart

## ASSESSMENT FEEDBACK REPORT

### FACILITATOR FEEDBACK & REMARKS

Well done

### ASSESSMENT JUDGEMENT

Learner's Total Mark:  
not met ☐

Requirements met ☒

Requirements

Action/s required:

N/A

By when:

### LEARNER FEEDBACK & COMMENTS

### DECLARATION BY THE FACILITATOR

I ..... (Facilitator) hereby certify that I have examined the learner workbook and I am satisfied with the evidence provided by the learner.

### Application for Assessment

First Attempt ☐ Second Attempt ☐ Third Attempt ☐ RPL ☐

Observer/Witness required: Yes/No

Name: \_\_\_\_\_

Contact Number: \_\_\_\_\_

Interpreter required: Yes/No

Name: \_\_\_\_\_

Contact Number: \_\_\_\_\_

Was Employer informed of request for Assessment: Yes/No

Type of Assessment:      Theoretical  
                                         Practical

### Recommendation for Assessment

The applicant has completed his/her training, has prepared for the assessment and is now ready to be assessed.

The ETDP Practitioner believe that the candidate is ready for the assessment because he has completed the relevant unit standard.

### Assessment Planning

|                                                 | <u>Theory</u> | <u>Practical</u> |
|-------------------------------------------------|---------------|------------------|
| Date of assessment:                             |               |                  |
| When would be the best time for the Assessment? | 9H00          | 9H00             |

Have you received an application to assess the candidate? Yes

What assessment methods do you consider using? (observation, assignment, presentation, etc.) Explain method.

1. Written Assignment : Printed Questionnaire
2. Observation : As per Practical Assessment Guide
3. Interview : List of Questions as per Practical Assessment Guide after Practical Observation

What resources would you require to conduct the assessment?

1. Assessor toolkit
2. Theoretical assignment
3. Practical assessment
4. Learner toolkit

How would you sequence your activities? Explain

1. Record preparation process
2. Meet with group/candidate and explain assessment procedure – how you are going to conduct the assessment
3. Learner will write summative assessments